

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks Available |     |     |       |       |      |
|----------|-----|------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                   | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 4.<br>HT | (a) |      |  | Fertilisers {increase the yield/ make plants grow<br>{quicker/bigger/more}/prevent deficiency diseases} (1)<br>Fertilisers can pollute waterways/ cause weed growth / cause eutrophication (1)<br><br>Pesticides kill pests (1)<br>Pesticides can build up in food chain/ harm top consumer/ kills non-target species / bioaccumulation / forms resistance in pests (1)                                                                           | 4               |     |     | 4     |       |      |
|          | (b) | (i)  |  | $68 - 29 = 39$ (1)<br>$39/5$ (1)<br>$= 7.8$ (1)<br><br>OR<br><br>$29 \div 5 = 5.8$ (1)<br>$68 \div 5 = 13.6$ (1)<br>Difference= $7.8$ (mm/week) (1)                                                                                                                                                                                                                                                                                               |                 | 3   |     | 3     | 2     |      |
|          |     | (ii) |  | 0.2 to 0.4 increase from 29 to 58 / $58 \div 29 = 2$ (1)<br>0.4 to 0.8 increase from 58 to 73 / $73 \div 58 = 1.2$ / should increase for 58 to 116 (1)<br>so do not agree (1)<br>Do not allow 3rd marking point without justification<br><br>For 2 marks:<br>Incomplete use of data in agreement and disagreement followed by 'do not agree'<br><br>For 1 mark:<br>Incomplete use of data in agreement or disagreement followed by 'do not agree' |                 |     | 3   | 3     | 3     |      |

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                         | Marks Available |          |          |           |          |      |
|----------|-----|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                         | AO1             | AO2      | AO3      | Total     | Maths    | Prac |
|          | (c) |  |  | <ul style="list-style-type: none"><li>• Selective breeding / selection of individuals with {favourable characteristics/largest rice yields} (1)</li><li>• Collect seeds from plants with largest rice yield / collect seeds from best plants (1)</li><li>• Cross {breeding/pollination} (of selected individuals) (1)</li><li>• Further selection of offspring / Repeat process over time (1)</li></ul> | 4               |          |          | 4         |          |      |
|          | (d) |  |  | Any 2 × (1) from:<br>Growing plants without soil / in water / in solution (1)<br>containing minerals/nutrients (1)<br>that is {pumped/moved} around (the plant roots) (1)                                                                                                                                                                                                                               | 2               |          |          | 2         |          |      |
|          |     |  |  | <b>Question 4 total</b>                                                                                                                                                                                                                                                                                                                                                                                 | <b>10</b>       | <b>3</b> | <b>3</b> | <b>16</b> | <b>5</b> |      |